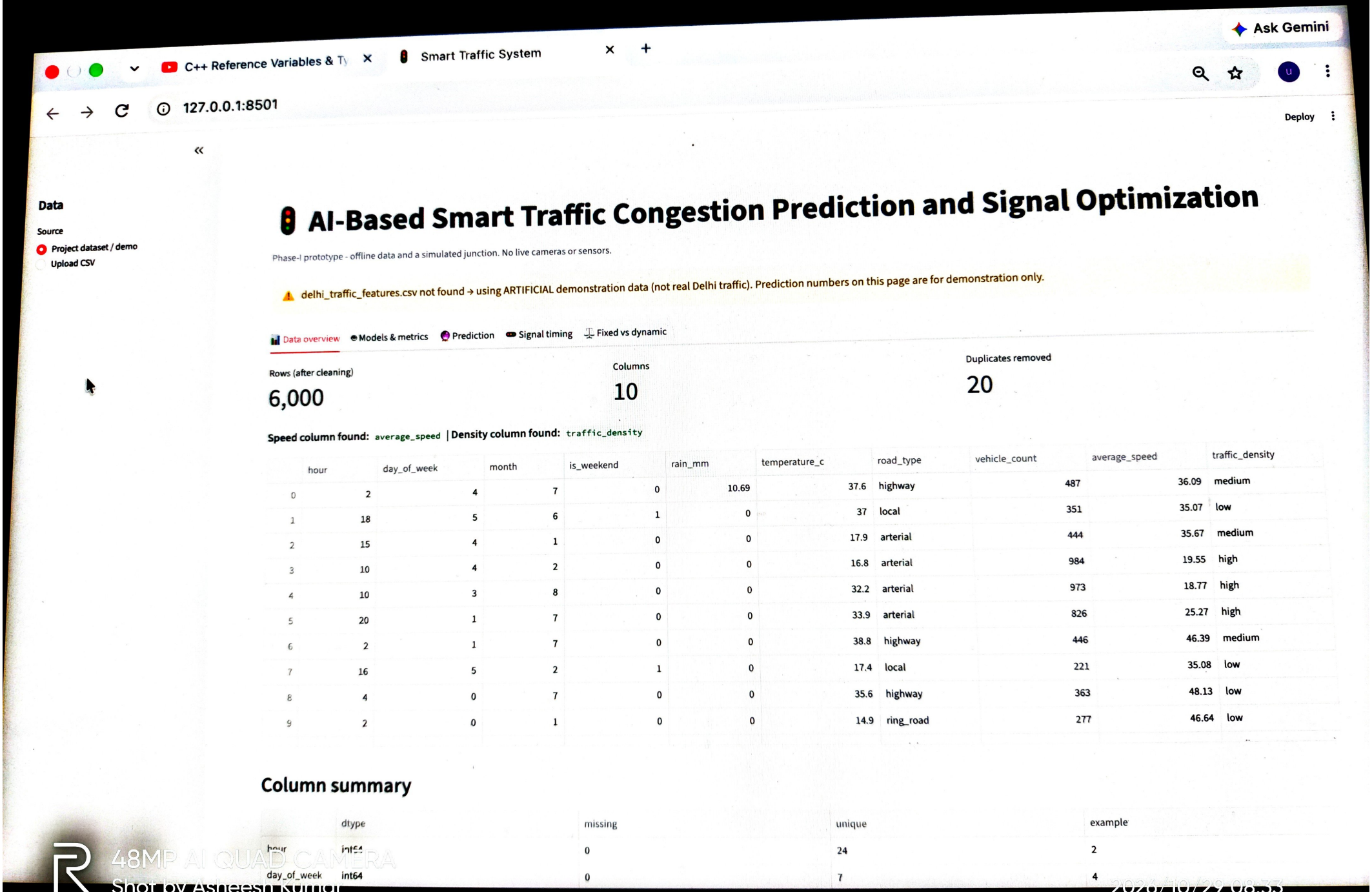


	hour	day_of_week	month	is_weekend	rain_mm	temperature_c	road_type	vehicle_count	average_speed	traffic_density
0	2	4	7	0	10.69	37.6	highway	487	36.09	medium
1	18	5	6	1	0	37	local	351	35.07	low
2	15	4	1	0	0	17.9	arterial	444	35.67	medium
3	10	4	2	0	0	16.8	arterial	984	19.55	high
4	10	3	8	0	0	32.2	arterial	973	18.77	high
5	20	1	7	0	0	33.9	arterial	826	25.27	high
6	2	1	7	0	0	38.8	highway	446	46.39	medium
7	16	5	2	1	0	17.4	local	221	35.08	low
8	4	0	7	0	0	35.6	highway	363	48.13	low
9	2	0	1	0	0	14.9	ring_road	277	46.64	low

	dtype	missing	unique	example
hour	int64	0	24	2
day_of_week	int64	0	7	4

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Deploy

Data

Source

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40	3	2	6	0	0	33.7	arterial	219	40.64	low
41	18	2	12	0	0	19.8	arterial	1046	19.88	high
42	16	2	7	0	0	31.8	highway	843	36.99	high
43	8	5	11	1	0	18.6	highway	1140	30.14	high
44	1	4	9	0	0	29	local	102	39.89	low
45	23	3	2	0	0	17.6	local	147	37.08	low
46	10	0	8	0	0	34.5	local	462	28.01	medium
47	21	3	11	0	0	22.8	local	322	33.86	low
48	16	1	2	0	12.1	23.1	arterial	755	12.91	high
49	18	2	11	0	0	20	arterial	1070	11.68	high

Column summary

	dtype	missing	unique	example
hour	int64	0	24	2
day_of_week	int64	0	7	4
month	int64	0	12	7
is_weekend	int64	0	2	0
rain_mm	float64	0	735	10.69
temperature_c	float64	60	284	37.6
road_type	str	60	4	highway
vehicle_count	int64	0	1376	487
average_speed	float64	0	2837	36.09
traffic_density	str	0	3	medium

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Deplo

AI-Based Smart Traffic Congestion Prediction and Signal Optimization

Phase-I prototype - offline data and a simulated junction. No live cameras or sensors.

Data

Source

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⚠ delhi_traffic_features.csv not found → using ARTIFICIAL demonstration data (not real Delhi traffic). Prediction numbers on this page are for demonstration only.

Data overview | Models & metrics | Prediction | Signal timing | Fixed vs dynamic

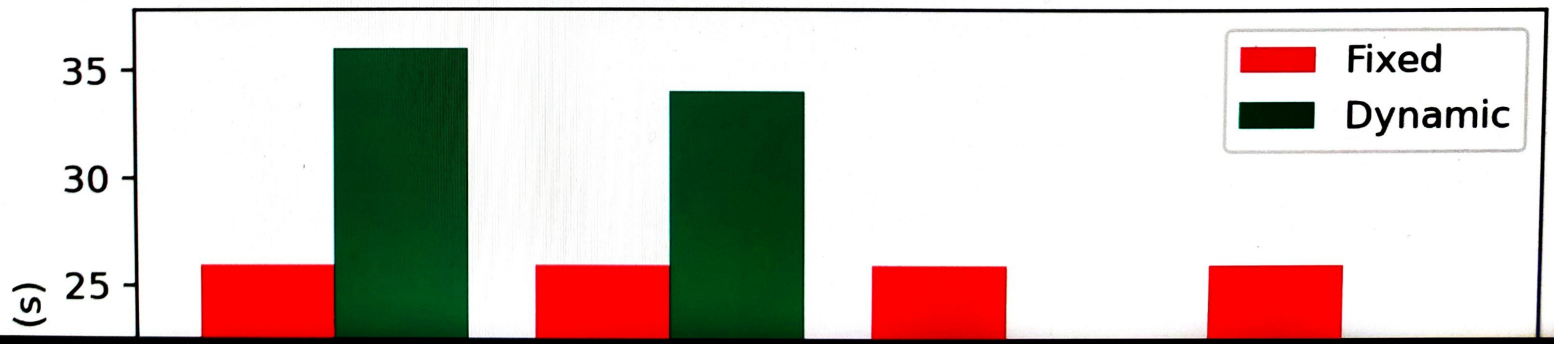
Share one signal cycle among four roads in proportion to demand. **Demand here is entered by you / simulated - the dataset has no per-road counts.**

Cycle (s)	Lost time / phase (s)	Min green (s)	Max green (s)
120	4	10	60

North demand (veh/h)	South demand (veh/h)	East demand (veh/h)	West demand (veh/h)
900.00	800.00	250.00	250.00

	Road	Demand (veh/h)	Fixed green (s)	Dynamic green (s)
0	North	900	26	36
1	South	800	26	34
2	East	250	26	17
3	West	250	26	17

Green 104 s + lost 16 s = 120 s = cycle 120 s ✓



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3 West 250 26 17

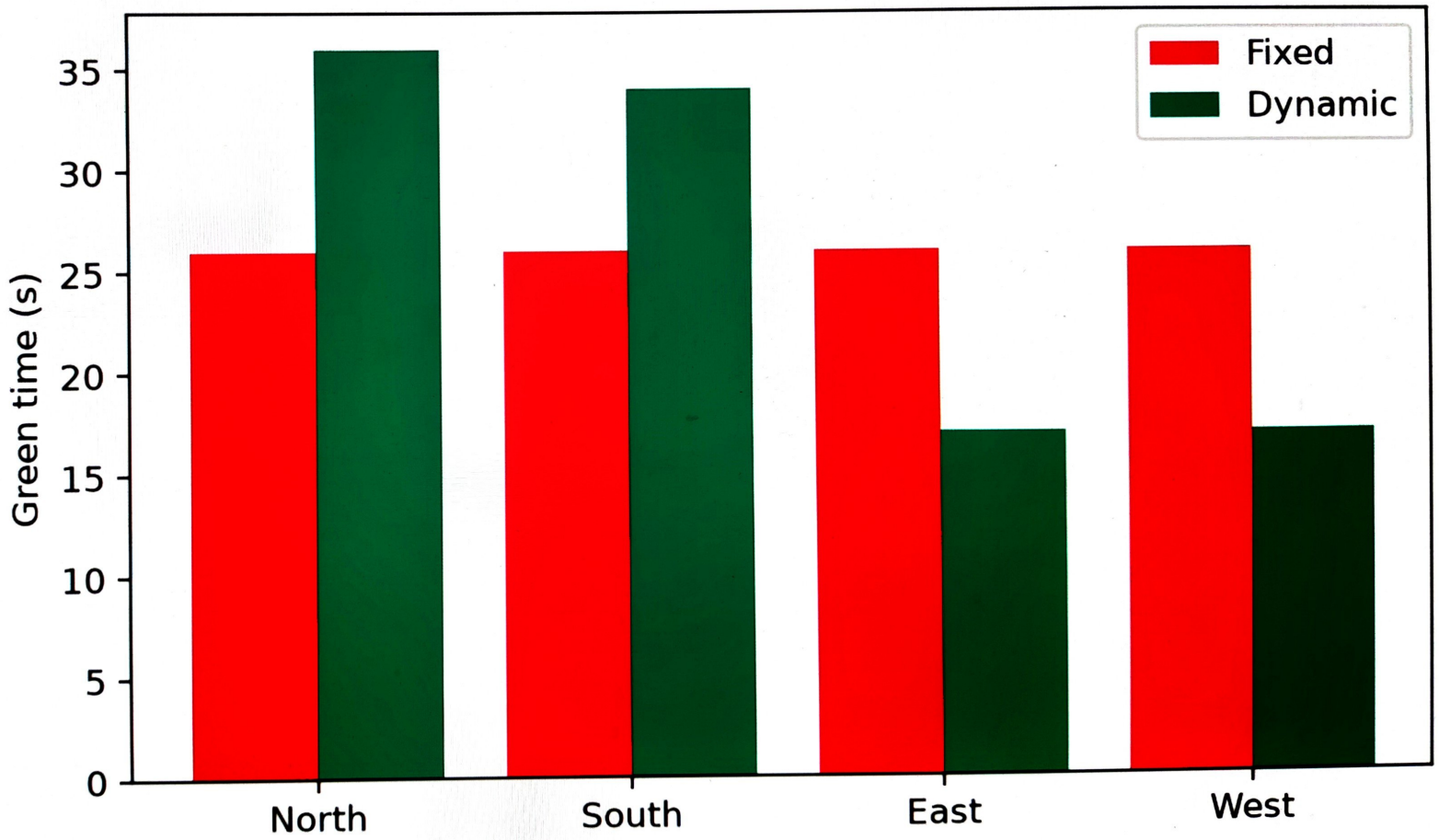
Data

Source

Project dataset / demo

Upload CSV

Green 104 s + lost 16 s = 120 s = cycle 120 s ✓



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Data

Source

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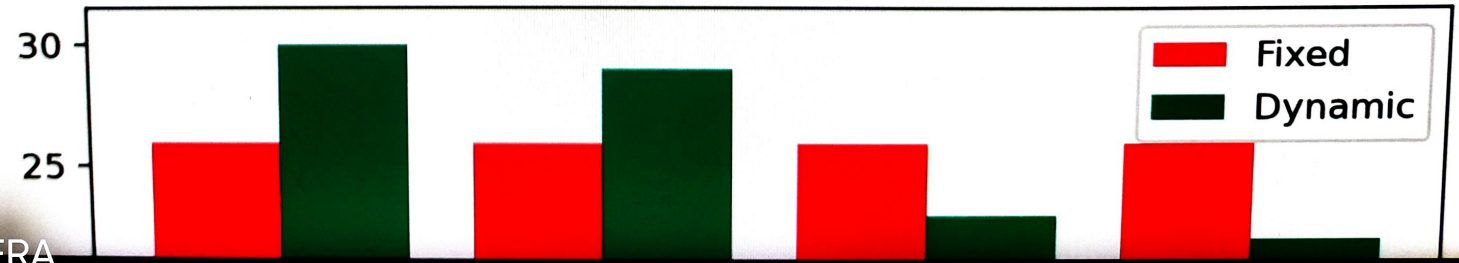
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120	4	20	60

North demand (veh/h)	South demand (veh/h)	East demand (veh/h)	West demand (veh/h)
900.00	800.00	250.00	250.00

	Road	Demand (veh/h)	Fixed green (s)	Dynamic green (s)
0	North	900	26	30
1	South	800	26	29
2	East	250	26	23
3	West	250	26	22

Green 104 s + lost 16 s = 120 s = cycle 120 s ✓



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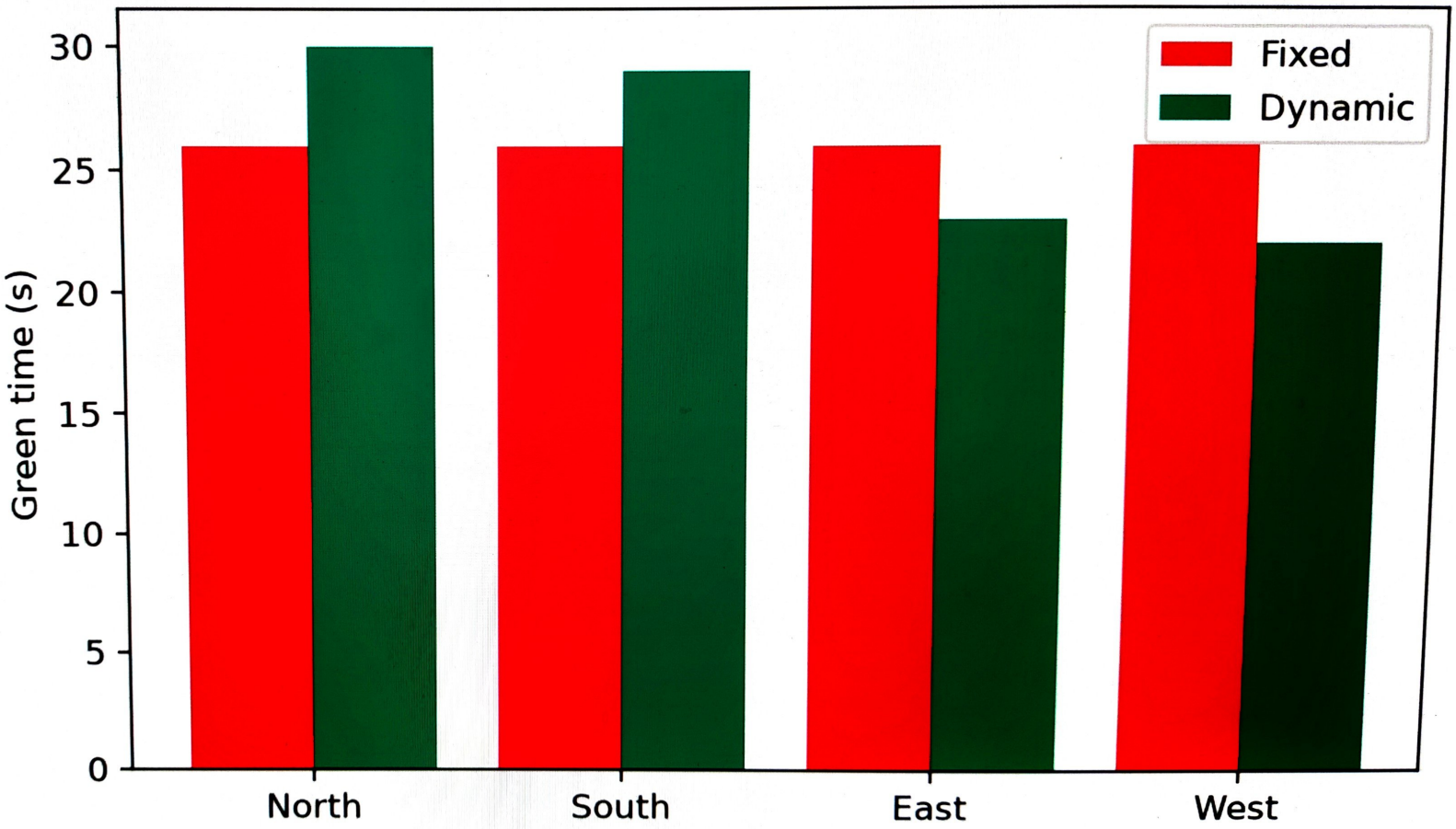
3	West	250	26	22

Data

Source

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Green 104 s + lost 16 s = 120 s = cycle 120 s ✓



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Data

Source

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Upload CSV

Phase-I prototype - offline data and a simulated junction. No live cameras or sensors.

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Data overview | Models & metrics | Prediction | Signal timing | Fixed vs dynamic

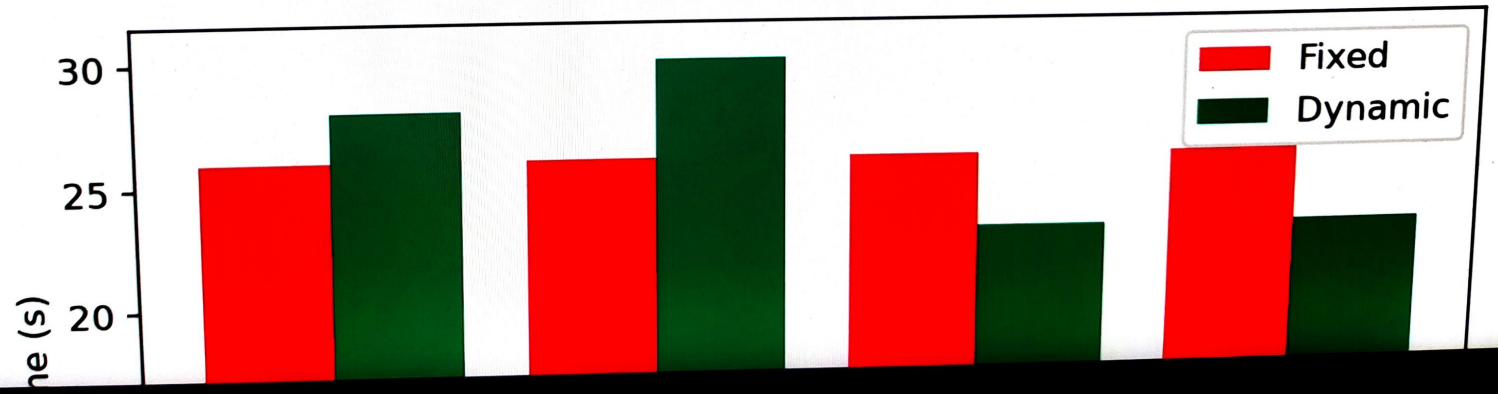
Share one signal cycle among four roads in proportion to demand. **Demand here is entered by you / simulated - the dataset has no per-road counts.**

Cycle (s)	Lost time / phase (s)	Min green (s)	Max green (s)
120	4	20	60

North demand (veh/h)	South demand (veh/h)	East demand (veh/h)	West demand (veh/h)
650.00	800.00	250.00	250.00

	Road	Demand (veh/h)	Fixed green (s)	Dynamic green (s)
0	North	650	26	28
1	South	800	26	30
2	East	250	26	23
3	West	250	26	23

Green 104 s + lost 16 s = 120 s = cycle 120 s ✓



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3 West 250 26 23

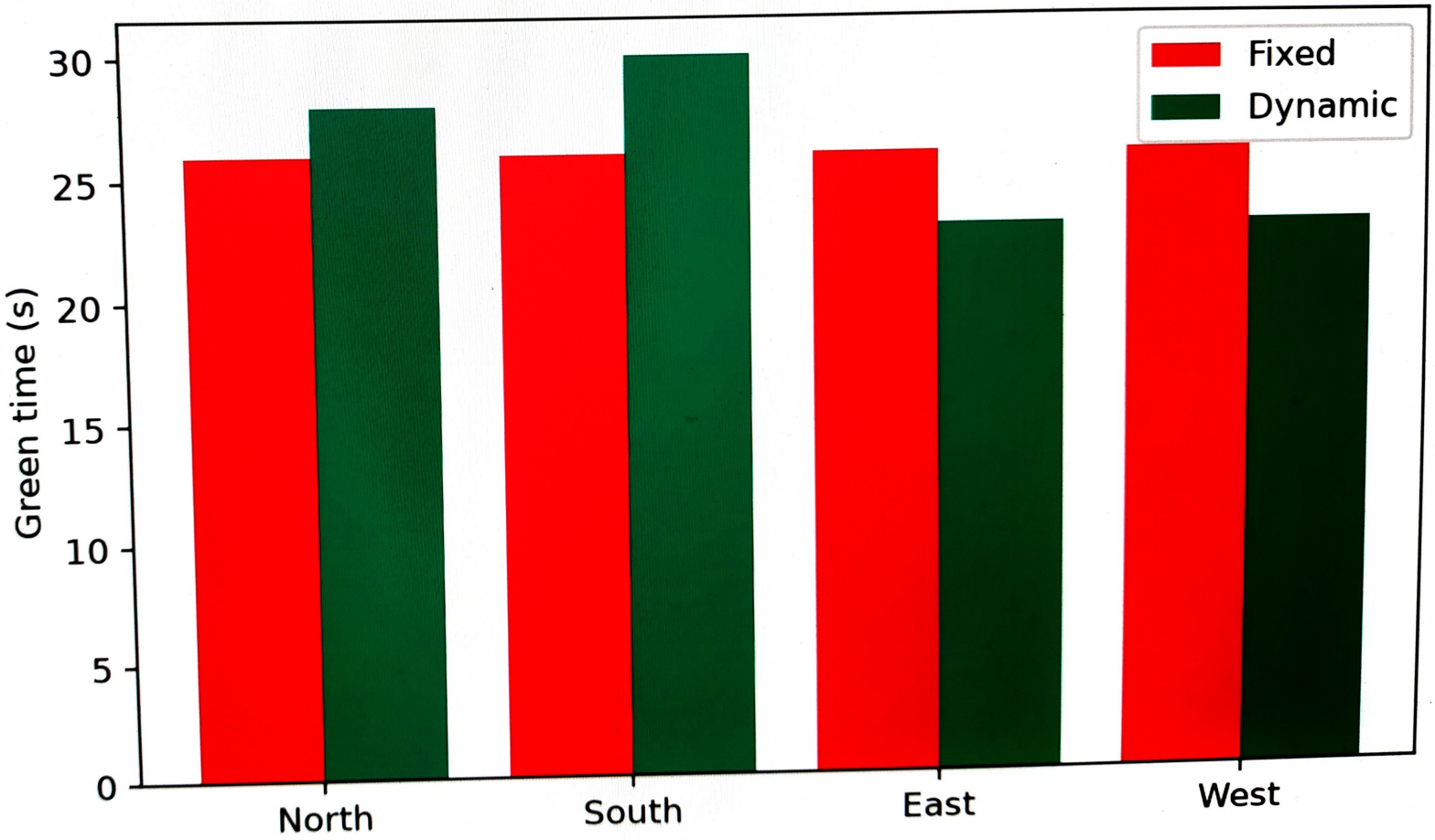
Data

Source

Project dataset / demo

Upload CSV

Green 104 s + lost 16 s = 120 s = cycle 120 s ✓



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